# SEL RTAC

Industrial Control Systems · SCADA Hub

## COURSE DESCRIPTION

The SEL Real-Time Automation Controller is the standard platform for substation SCADA integration, protocol translation, and local logic execution in power utility environments. This course covers hardware selection, ACSELERATOR Architect project configuration, DNP3 and Modbus gateway setup, IEC 61850 service integration, IEC 61131-3 logic programming, and NERC CIP access control requirements.

## PREREQUISITES

Completion of the DNP3 and IEC 61131-3 courses. Basic familiarity with substation equipment is helpful.

## COURSE TOPICS

1. **RTAC Architecture:** SEL-3505/3530/3555 hardware variants, I/O module selection, communication ports, and physical commissioning steps.
2. **ACSELERATOR Architect:** Project database structure, tags, aliases, protocol point mapping, revision control, and first-live-data workflow.
3. **DNP3 Configuration:** Outstation and master roles, point database mapping, event class assignment, unsolicited reporting, and SCADA latency parameters.
4. **Modbus Gateway:** TCP server configuration, DNP3-to-Modbus point mapping, register layout, data type conversion, and translation failure modes.
5. **IEC 61850 Integration:** GOOSE trip signals, Sampled Values, MMS operator control, and internal tag database mapping for each service type.
6. **Local Logic:** IEC 61131-3 programs, scan cycle behavior, event-driven automation, and design principles for WAN-independent operation.
7. **Security and Access Control:** Role-based access, serial port security, firmware update procedures, and NERC CIP implications per configuration decision.
8. **Field Scenarios:** Three case studies — automatic tap changer response, feeder protection with downstream reclosers, and sub-200ms load shedding.

## LEARNING OUTCOMES

- Commission an SEL RTAC from hardware selection through first live DNP3 tag.
- Map DNP3, Modbus, and IEC 61850 protocol points to the RTAC internal tag database.
- Write IEC 61131-3 local logic programs that operate correctly without WAN connectivity.
- Configure role-based access control and document each decision against NERC CIP requirements.
- Analyze and implement field automation scenarios including tap changer and load shedding sequences.

## REQUIRED SOFTWARE

- **ACSELERATOR Architect** (free from selinc.com, Windows) — SEL's project configuration tool for all RTAC lab exercises.
- **SEL-5033 Software Defined Network** (free, Windows) — virtual RTAC environment for lab scenarios without physical hardware.